

Create Job plans in Maximo from Asset maintenance manuals.

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1. Background

Asset-intensive industries often rely on **OEM(Original Equipment Manufacturer)-provided maintenance manuals** to guide equipment upkeep. These manuals typically contain unstructured, narrative-style instructions that are difficult to operationalize directly within enterprise systems like **IBM Maximo**.

Traditionally, maintenance engineers manually interpret these manuals and input tasks into Maximo as job plans—a process that is **time-consuming, error-prone, and non-scalable**. This bottleneck delays maintenance planning, introduces inconsistencies across teams, and increases the risk of non-compliance with OEM standards.

To overcome these challenges, this use case proposes an **AI-powered solution** that automates the conversion of maintenance manual content into **structured Maximo job plans**.

Using IBM Watsonx capabilities:

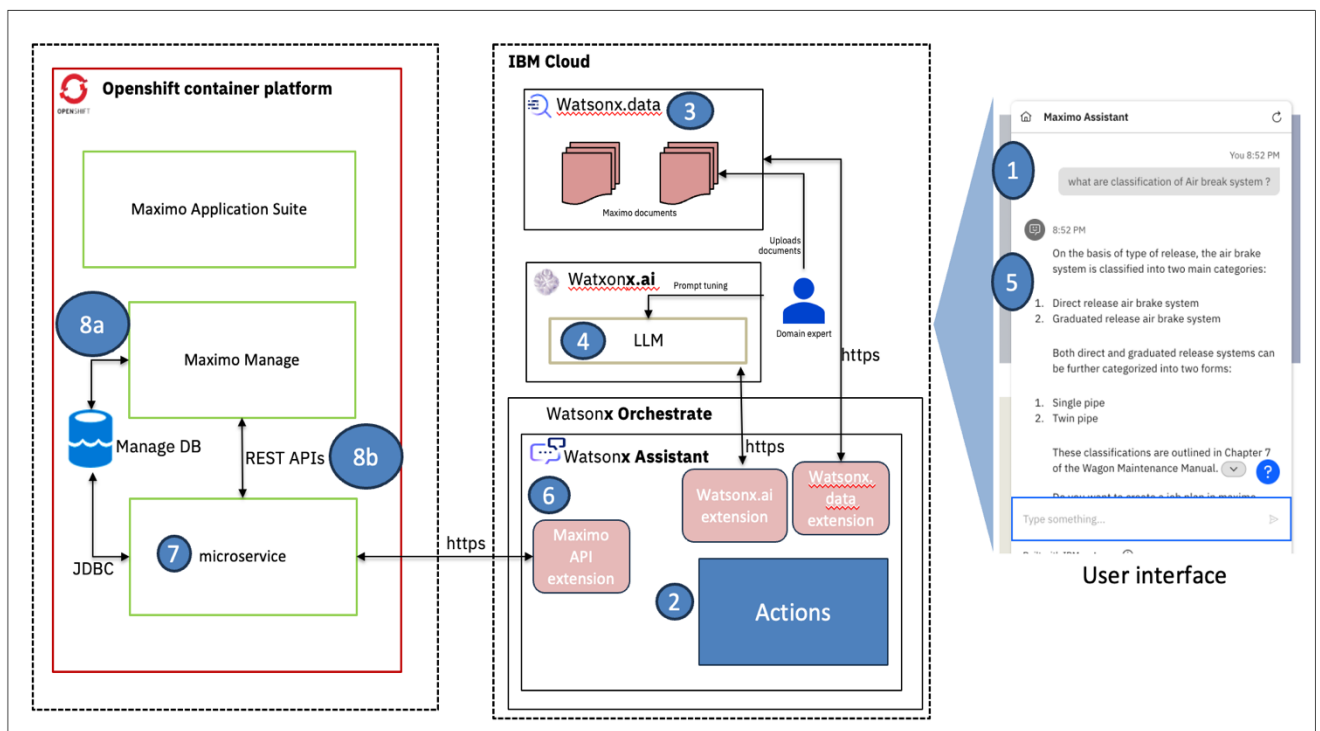
- **Watsonx.ai** is leveraged to interpret and extract task sequences from unstructured PDF manuals using large language models (LLMs).
- The extracted steps are transformed into a JSON structure compatible with Maximo's job plan schema.
- These plans are then ingested into Maximo via APIs, enabling **automated creation and updates of job plans**.
- **Watsonx.data** (with Milvus) supports intelligent retrieval and indexing of manual content for RAG (Retrieval Augmented Generation) scenarios.

This automated pipeline ensures:

- **Standardized job plan generation**
- **Reduced manual effort**
- **Faster turnaround for preventive and corrective maintenance planning**
- **Alignment with OEM-specified procedures**

By streamlining this process, your maintenance team can scale efficiently, improve data integrity, and minimize equipment downtime

2. Architecture Diagram



3. Technologies used

Services	Purpose
Watsonx.ai Studio	To generate structured responses using LLMs
Watsonx.ai Runtime	To host and run foundation models
Watsonx Assistant	For building conversational search applications
Watsonx.Data (Milvus DB)	Storing and retrieving vectorized document data
Maximo	Asset Management Tool

4. Benefits Milvus Database

We are extracting and generating **job plans from asset maintenance manuals** using AI. These manuals are often large, unstructured documents with complex procedures scattered across many pages. To accurately retrieve relevant maintenance steps in real time, we need a **high-performance vector database** that can:

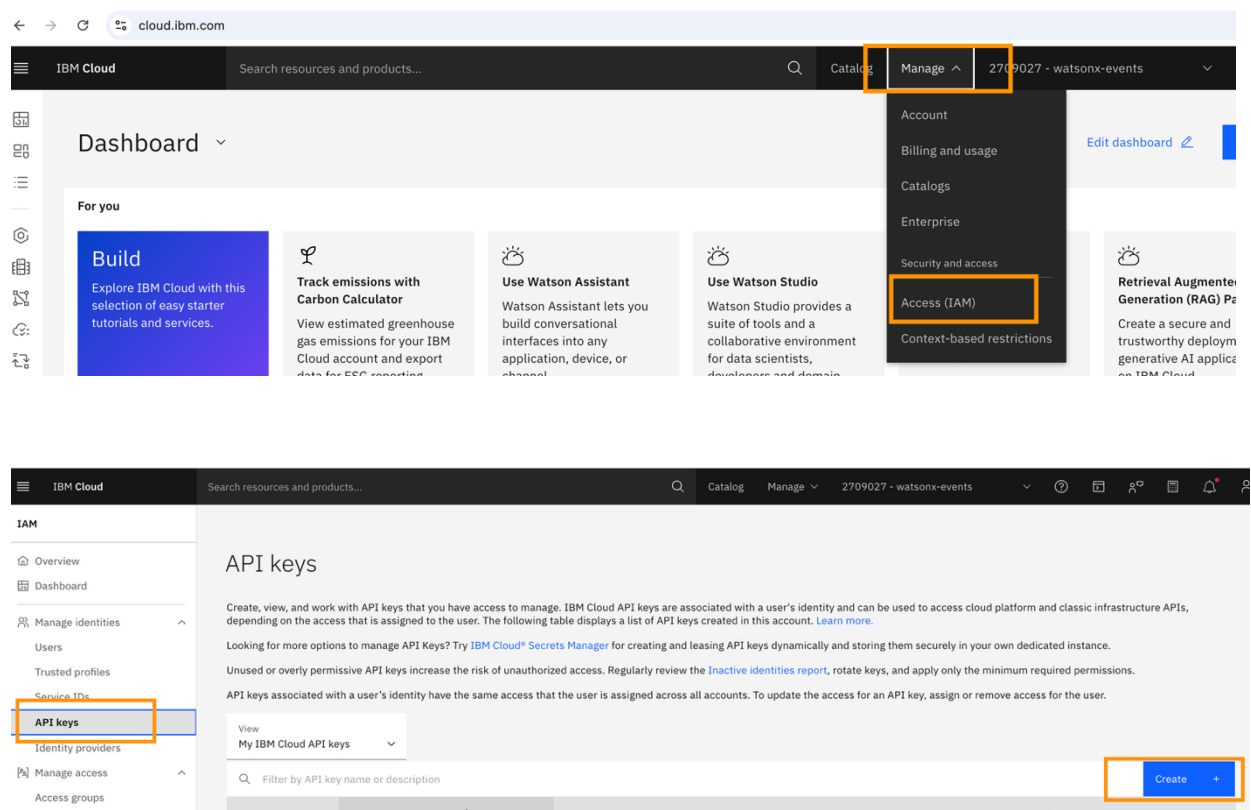
Feature	Why It Matters in This Use Case
 Vector Similarity Search	Enables retrieval of the most relevant content chunks from manuals based on semantic similarity to the user's question or task.
 High Performance	Designed for large-scale vector data, Milvus ensures fast retrieval even with thousands of embedded manual pages.
 RAG Enablement	Powers the Retrieval Augmented Generation (RAG) workflow by feeding only the most relevant context to Watsonx.ai for accurate and grounded responses.
 Unstructured Data Indexing	Stores and indexes large volumes of text embeddings from PDFs, maintenance guides, and documents.
 Seamless Integration	Works well with Watsonx.data and is supported within the IBM Cloud environment via infrastructure manager.

5. Create API key

Why to Create API key: An API key is required to securely authenticate and authorize communication between services in your Watsonx and Maximo environment.

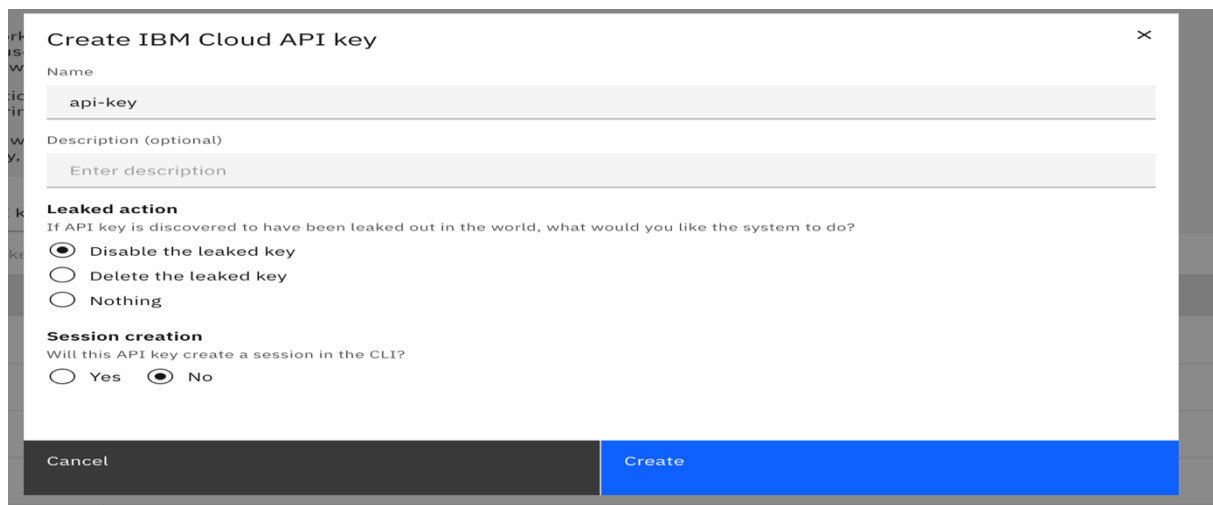
- Always keep API keys confidential.
- Rotate them periodically.
- Never hard-code them directly into source code (use .env or secrets managers instead).

5.1 Go to **IBM Cloud** dashboard → click on **Manage** → click on **Access (IAM)** → then click on **API Keys** on the left side → click on **Create**



The image shows two screenshots of the IBM Cloud interface. The top screenshot shows the 'Manage' dropdown menu with 'Access (IAM)' highlighted. The bottom screenshot shows the 'API keys' page with the 'Create' button highlighted.

5.2 Provide *Name* and *Description* → Click on *Create* → Copy & download the API key for future reference



Create IBM Cloud API key

Name
api-key

Description (optional)
Enter description

Leaked action
If API key is discovered to have been leaked out in the world, what would you like the system to do?
☒ Disable the leaked key
☐ Delete the leaked key
☐ Nothing

Session creation
Will this API key create a session in the CLI?
☐ Yes ☒ No

Cancel Create

6. Watsonx.data Configurations Details

6.1 Copy the below connection details provided by the instructor.

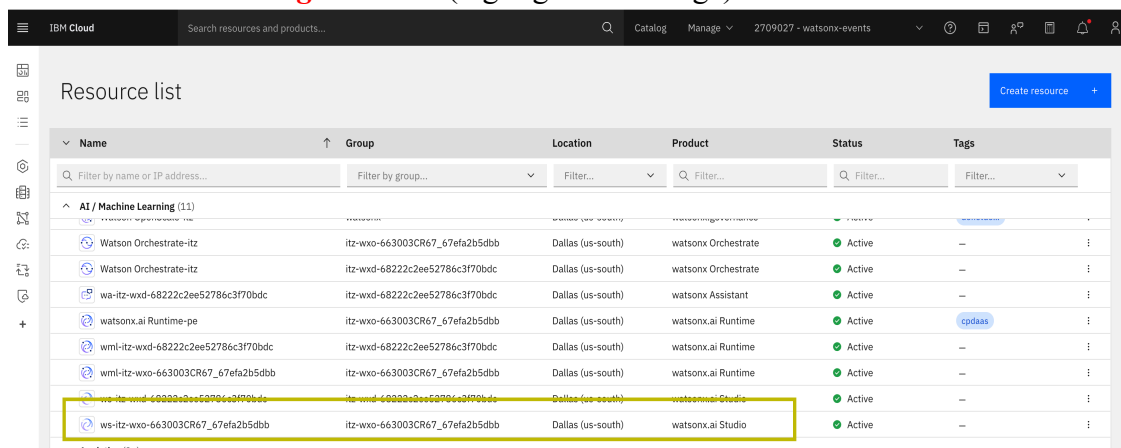
GRPC host: <host name>

GRPC port: <port>

7. Watsonx.AI Configurations

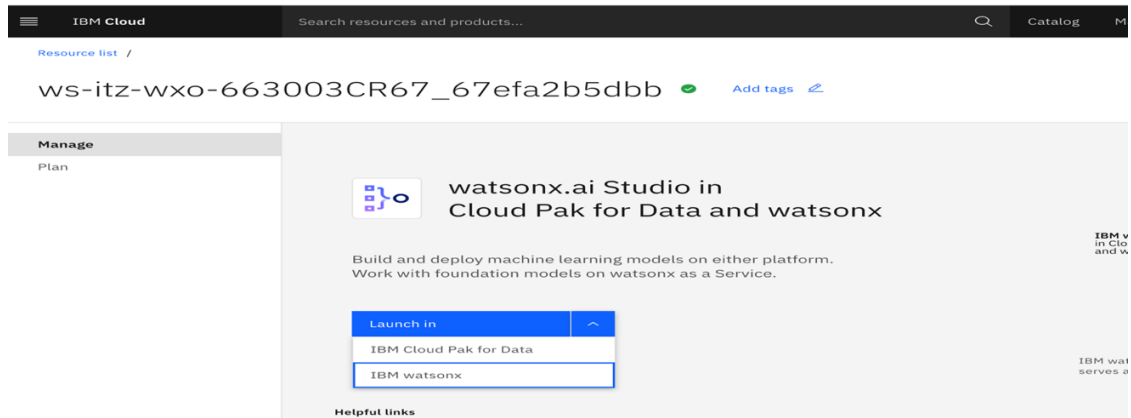
7.1 Launch watsonx.ai

Click on *Resource list* from left side panel in *IBM Cloud* → Click on *watsonx.ai Studio* under *AI / Machine Learning* resource. (highlighted in image)



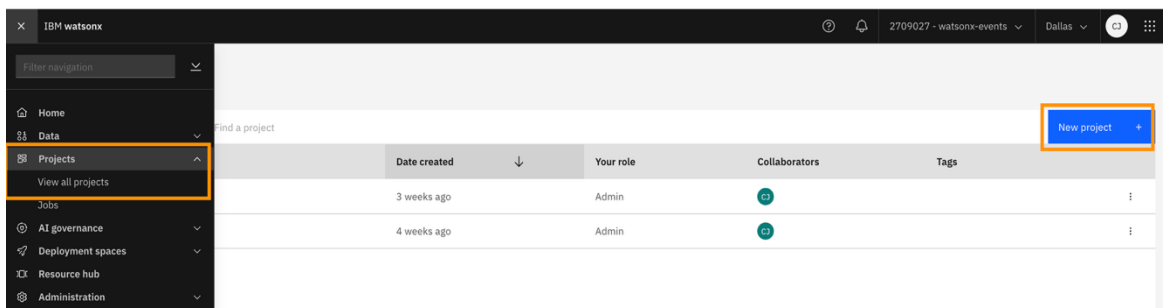
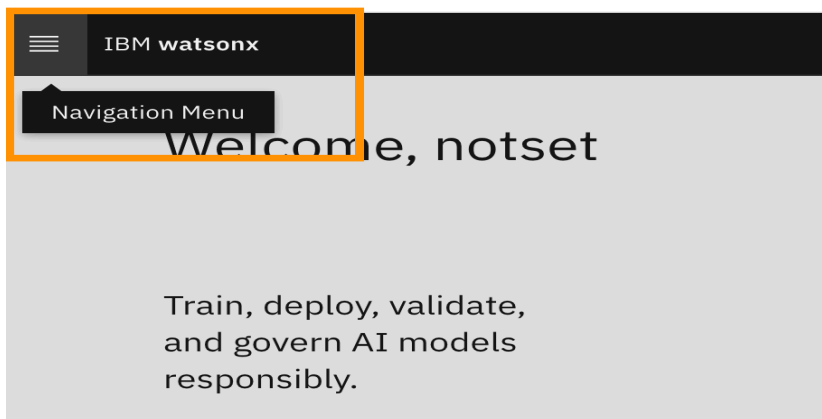
Name	Group	Location	Product	Status	Tags
AI / Machine Learning (11)					
Watson Orchestrate-itz	itz-wxo-663003CR67_67efa2b5dbb	Dallas (us-south)	watsonx Orchestrate	Active	
Watson Orchestrate-itz	itz-wxd-68222c2ee52786c3f70bdc	Dallas (us-south)	watsonx Orchestrate	Active	
wa-itz-wxd-68222c2ee52786c3f70bdc	itz-wxd-68222c2ee52786c3f70bdc	Dallas (us-south)	watsonx Assistant	Active	
watsonx.ai Runtime-pe	itz-wxo-663003CR67_67efa2b5dbb	Dallas (us-south)	watsonx.ai Runtime	Active	cpdaas
wmi-itz-wxd-68222c2ee52786c3f70bdc	itz-wxd-68222c2ee52786c3f70bdc	Dallas (us-south)	watsonx.ai Runtime	Active	
wmi-itz-wxo-663003CR67_67efa2b5dbb	itz-wxo-663003CR67_67efa2b5dbb	Dallas (us-south)	watsonx.ai Runtime	Active	
ws-itz-wxd-68222c2ee52786c3f70bdc	itz-wxd-68222c2ee52786c3f70bdc	Dallas (us-south)	watsonx.ai Studio	Active	
ws-itz-wxo-663003CR67_67efa2b5dbb	itz-wxo-663003CR67_67efa2b5dbb	Dallas (us-south)	watsonx.ai Studio	Active	

7.2 Select Launch in *IBM watsonx*



7.3 Create New project

Click on *Navigation Menu* at top left side → Click on *Projects* → Click on *View all projects* → Click on *New project*.



7.4 Provide *Name* and *Select storage service* from drop down → Click on *Create*.

Create a project

Start with a new, blank project or select from where to import an existing project.

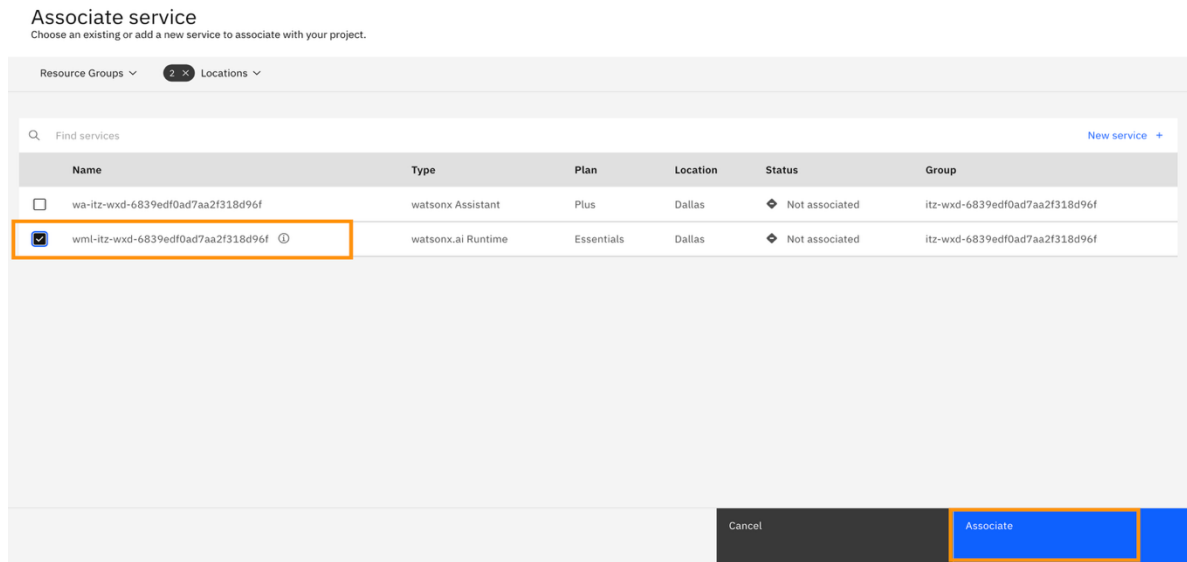
The screenshot shows the 'Create a project' form in IBM watsonx. On the left, there is a sidebar with options: '+ New', 'Local file', and 'Sample'. The main form has the following sections: 'Enter a name' (with 'project1' entered), 'Description (optional)' (with a text area), 'Tags (optional)' (with an 'Add tags' button and a note: 'Add tags to make projects easier to find. To add tags, separate them with commas and press Enter.'), 'Define storage' (with a green checkmark and 'Select storage service', a dropdown for 'Target Cloud Object Storage Instance' showing 'cos-itz-wxo-663003CR67_67efa2b5dbb', and a note: 'Project includes integration with Cloud Object Storage for storing project assets.'), and 'Advanced settings' (with a dropdown arrow). At the bottom right, there are 'Cancel' and 'Create' buttons.

The screenshot shows the IBM watsonx project overview page for 'project1'. The top navigation bar includes the IBM watsonx logo, a search bar, and user information. The main navigation tabs are 'Overview', 'Assets', 'Jobs', and 'Manage'. The 'Overview' tab is active, showing a 'Start working' section with four recommended actions: 'Add users as collaborators', 'Add data to work with', 'Chat and build prompts with foundation models', and 'Tune a foundation model with labeled data'. Below this, there are three panels: 'Jump back in' (showing assets), 'Resource usage' (showing '0 CUH' and '0 Tokens'), and 'Your documentation' (with a 'New!' badge and an 'Open Documentation editor' button).

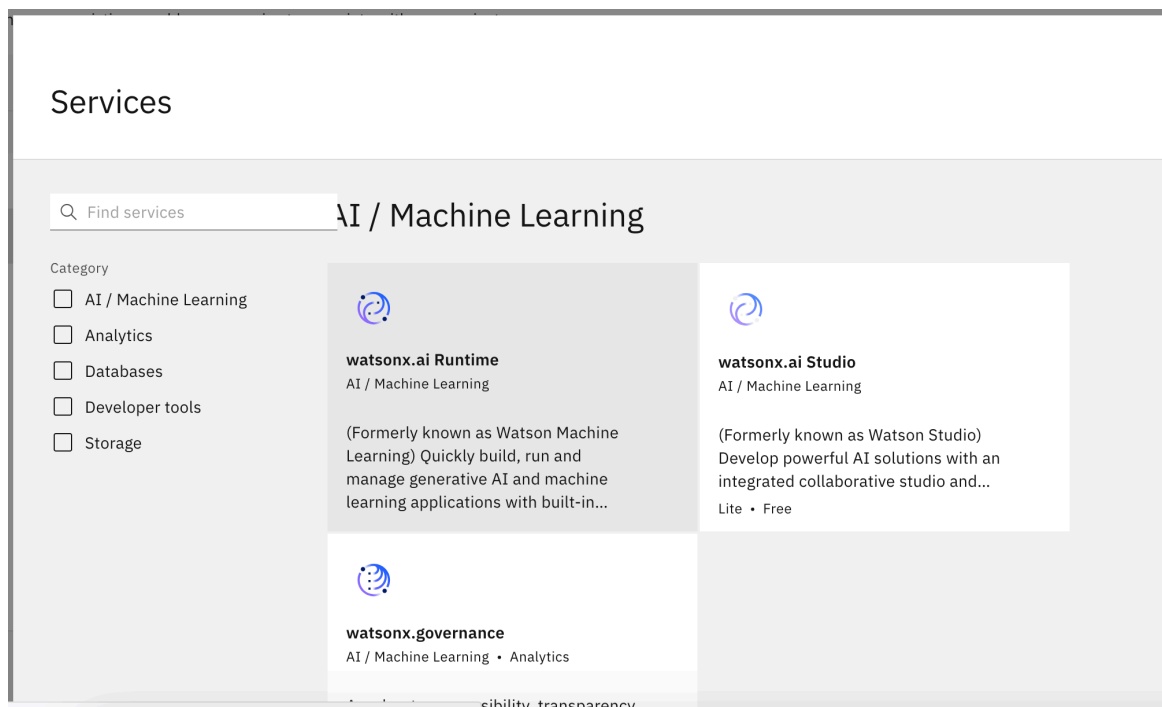
7.5 Once project created, Navigate to *Manage* tab → select *Services & integrations* (left side panel) → click on *Associate service* (highlighted in image)

The screenshot shows the 'Manage' tab in IBM watsonx, specifically the 'Services & integrations' section. The left sidebar has a 'Services & integrations' option highlighted with a blue box. The main content area has two tabs: 'IBM services' (active) and 'Third-party integrations'. Below the tabs, there is a text block: 'Associate IBM Cloud services with this project to add tools, compute environments, or other capabilities. [Learn more.](#)'. There is a search bar labeled 'Find services'. Below the search bar, there is a table with columns 'Name' and 'Service type'. At the bottom right, there is a blue button labeled 'Associate service' with a plus icon, highlighted with a blue box.

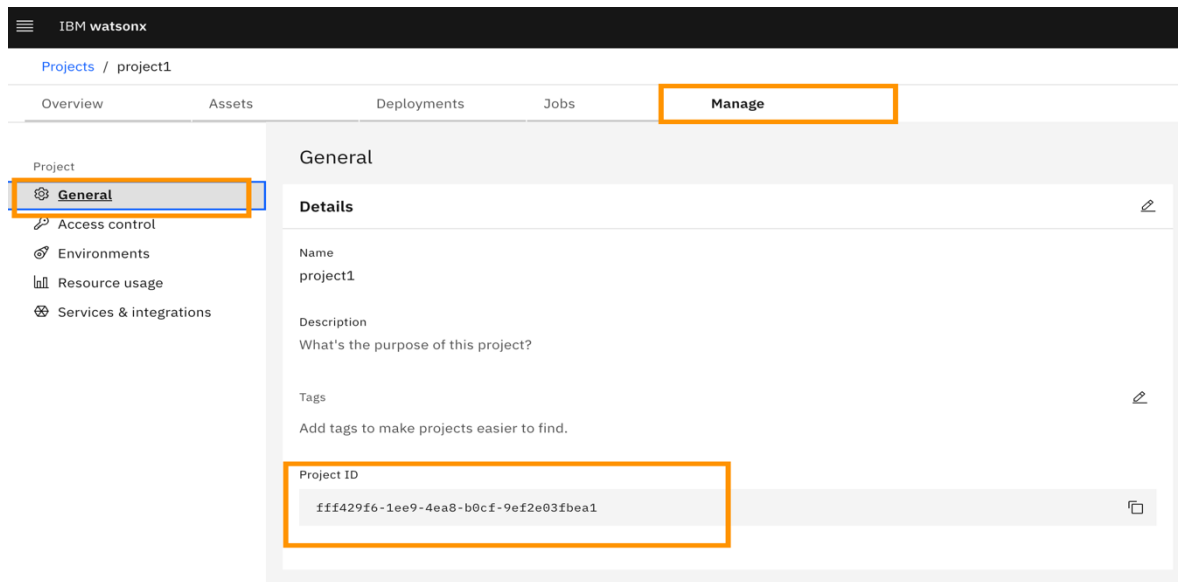
7.6 Select service name starts with wml(ex: wml-xxx-xxx-xxxxxxx) → click on *Associate*.



Note: if wml-xxx service does not display. Then click on New Service + icon and then select watsonx.ai Runtime as shown below and click on create.



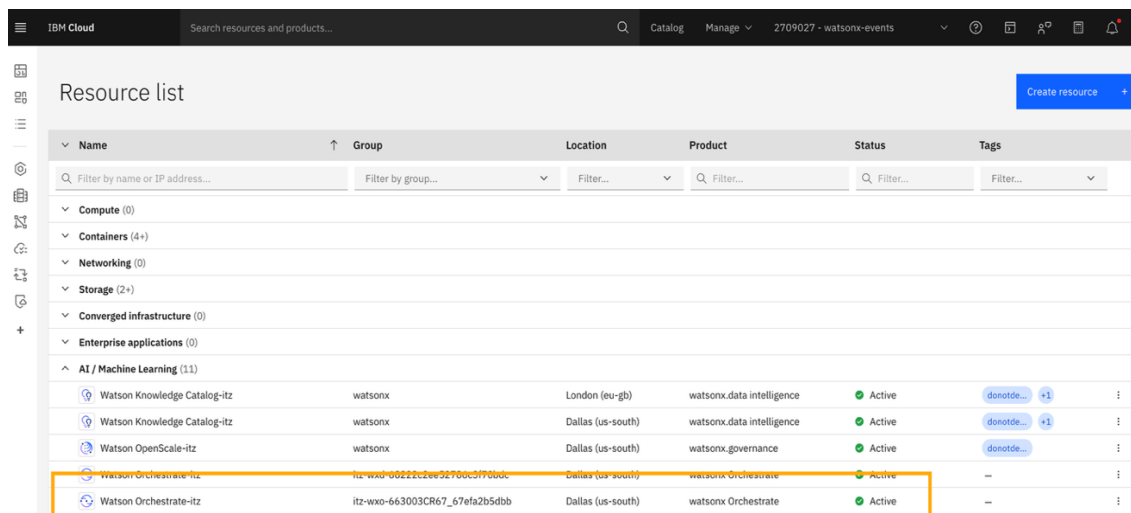
7.7 Again on *Manage* tab → Click on *General* → Copy and Save *Project ID* for the future use



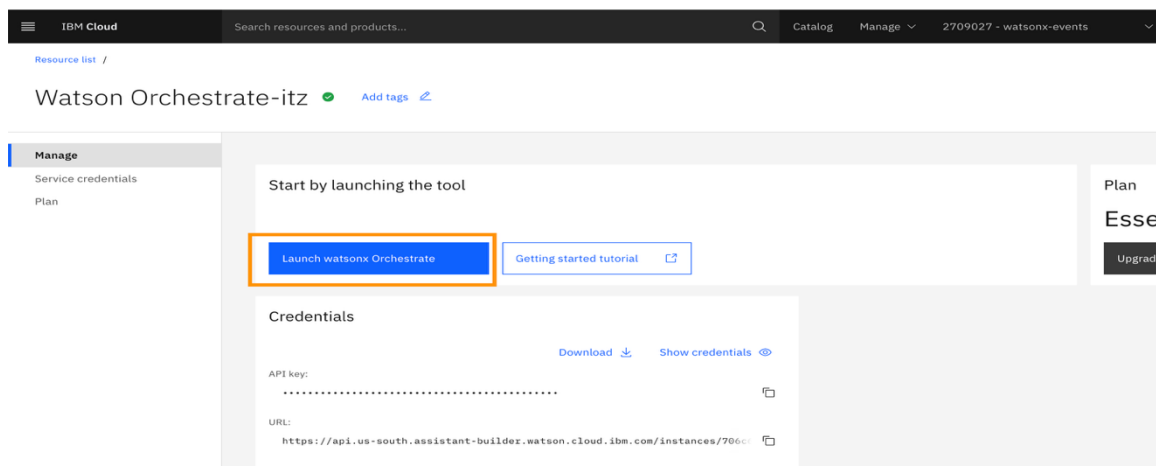
8. Watsonx Assistant Configurations

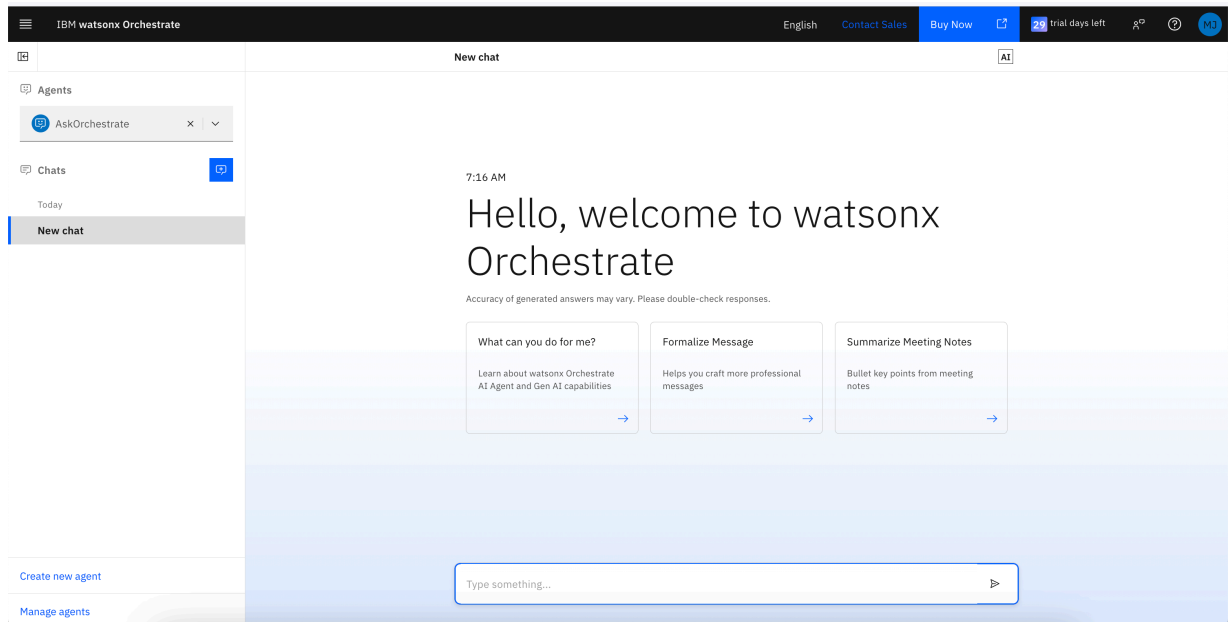
8.1 Launch watsonx assistance from orchestrate

Go to Resource list in **IBM Cloud** → expand **AI/Machine Learning** → Click on **Watson Orchestrate**- (highlighted in image)

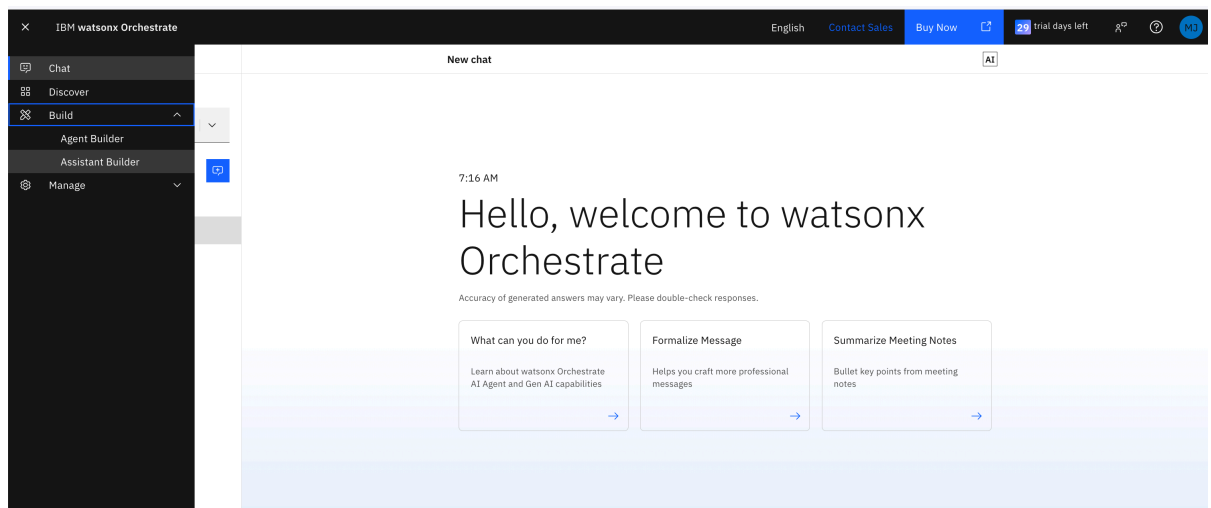


8.2 Click on **Launch watsonx Orchestrate**





8.3 Click on left menu, Expand *Build* and select *Assistant Builder* option



8.4 Enter the *Assistant Name* and optional Description and click on Next.

Note: Enter your initial name with Assistant Name, like Create-jobplan-XX

Welcome to AI assistant builder

Next

Create
Personalize
Customize
Preview

Create your first assistant

Let's get your assistant up and running. Name your assistant, add a description, and choose a language. In following steps we'll gather more information, show you basic customizations, and give you a preview of what your assistant will look like.

Assistant name

Your assistant name will be kept internally and not visible to your customers

Description (optional)

27/128

Create assistant page

Assistant language

English (US)

This is the language your assistant will speak.

8.5 Select the options as below to *personalize the Assistant*.

Note : You can select different values based on your requirement.

Welcome to AI assistant builder

BackNext

Create
Personalize
Customize
Preview

You may add multiple channels from your dashboard.

Where do you plan on deploying your assistant?

Web

Tell us about yourself

This information will be used to personalize your onboarding experience.

Which industry do you work in?

Utilities

What is your role on the team building the assistant?

Developer

Which statement describes your needs best?

I want to provide confident answers to common questions

Create assistant page

Do you have the Speed Demons in stock?

The Speed Demons are in stock at our Downtown and Northgate locations, which are both within 5 miles of you.

What size and color do you need?

I'm looking for a size 9 in white

Great news! The Speed Demons are available in white in a size 9.

You can purchase them for curbside pickup or we can ship them to you. Which would you prefer?

I'll pick them up!

Ship them to me!

8.6 Enter Assistant name as *Maximo Assistant* and feel free to select the different theme if you wish to.

Welcome to AI assistant builder

BackNext

Create
Personalize
Customize
Preview

Assistant's name as known by customers

Add an avatar image

Intended purpose

☒ Standard: For virtual agents and customer support experiences.
☐ Carbon for AI: For use in internal IBM products.

Choose a theme

Light

Dark

Primary color

Secondary color

Chat header

User message bubble

Accent color

Significant and interactive objects

Size

The size of the web chat on this page will not change by updating these fields.

Maximo Assistant

Hi! I'm a virtual assistant. How can I help you today?

Example: Find nearby location

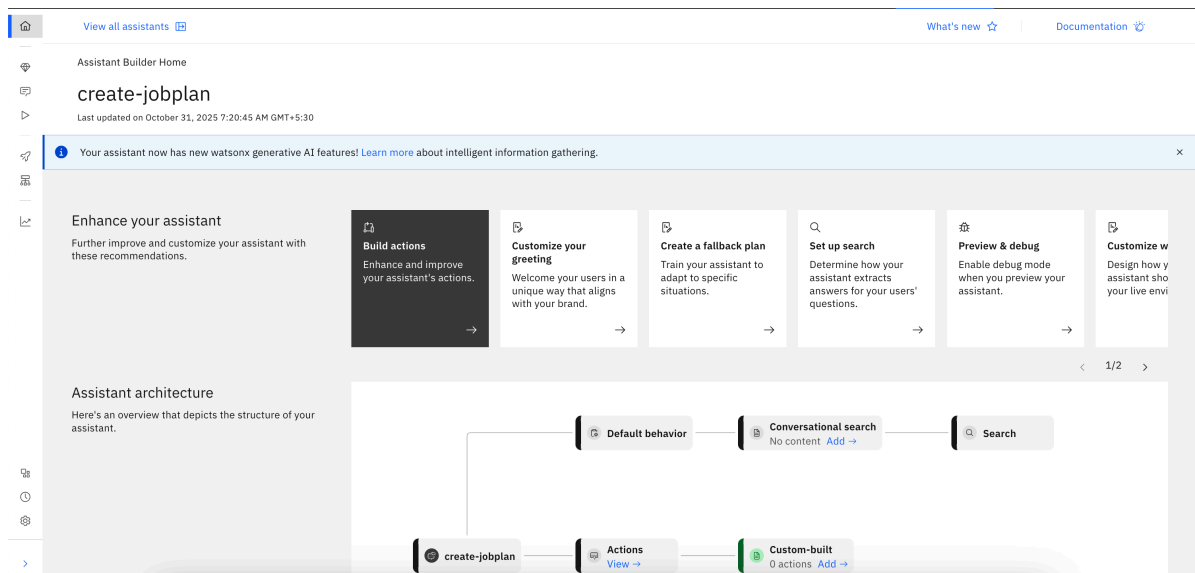
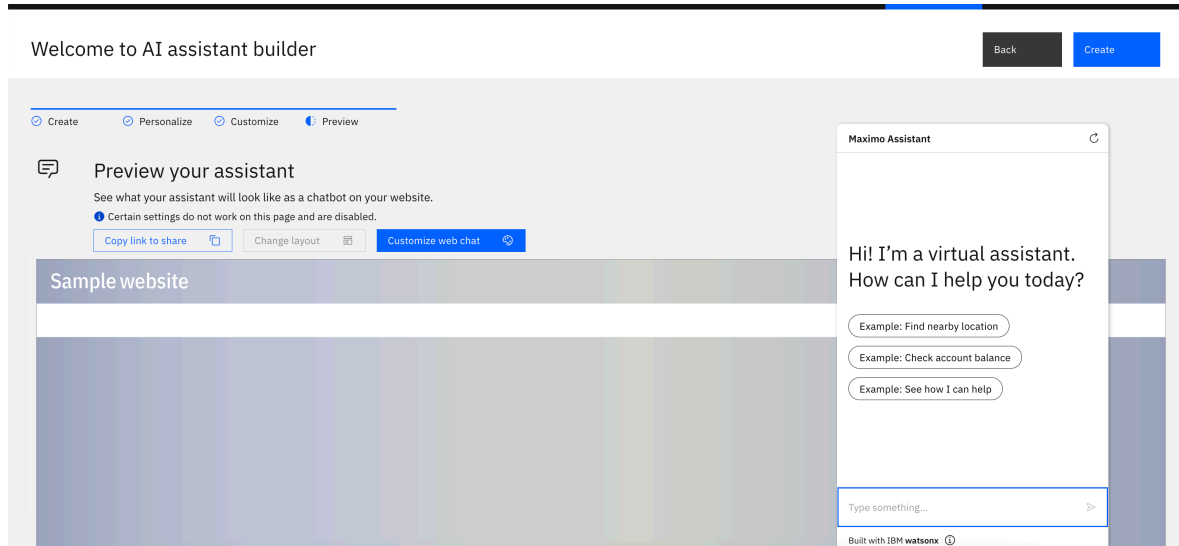
Example: Check account balance

Example: See how I can help

Type something...

11

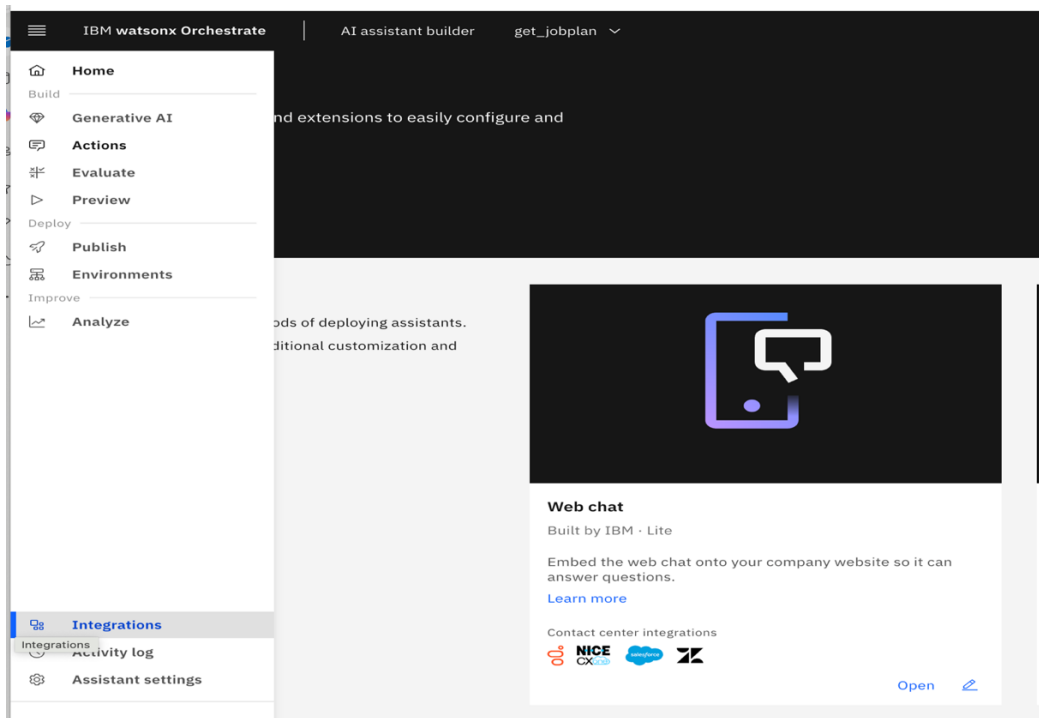
8.7 Click on Create button.



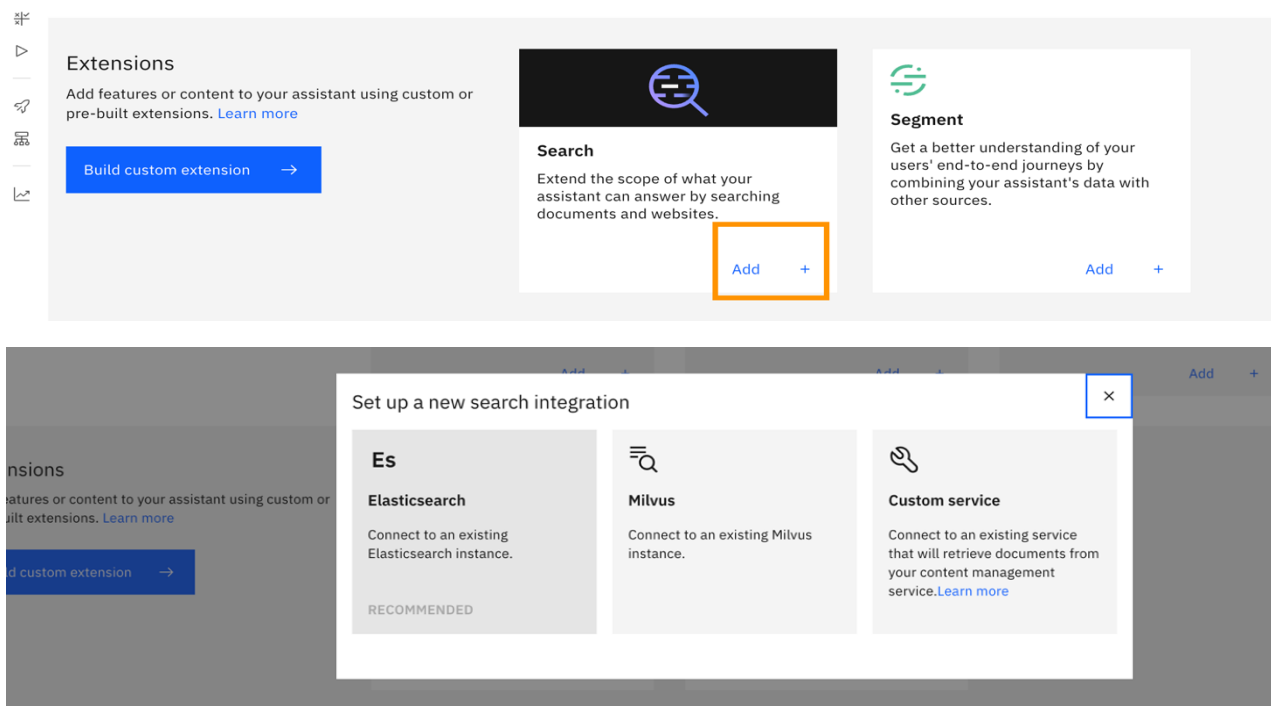
9. Adding Custom Extensions

9.1 Watsonx data extension (Milvus)

9.1.1 From Left Navigation, select *Integrations* option.



9.1.2 Look for *Search* → Click on *Add* → select *Milvus*



9.1.3 Provide Milvus service details and Click Next

GRPC host: → Copy from Startup.txt

GRPC port: → Copy from Startup.txt

API key: → Copy from Startup.txt

In case it shows to enter user name and password, use below details

Username : ibmlhapikey

Password: API Key

Connect Milvus

Select data source

Conversational search (opt

Connect your search provider

Fill in the information below to access your Milvus instance. [Learn more](#)

GRPC host

99e7d1a3-1f67-4252-8dd8-619ce4f92929.crkkivsd0vdrn0rdjn2g.lakehouse.appdomain.clou

GRPC port (required)

31384

Choose an authentication type

watsonx.data API key

API key

.....

9.1.4 Select *Database* → as Default

Collection name → asset_data

Choose index → embedding

Choose embedding_model_Id → all-minilm-l6-v2

Title -> file_name

Body -> text

Database

default

Choose collection

asset_data

Choose index

embedding

Choose embedding_model_id

all-minilm-l6-v2

Configure result content

Specify the Milvus fields you want to map to the title, body, and URL of the search response.

Title

file_name

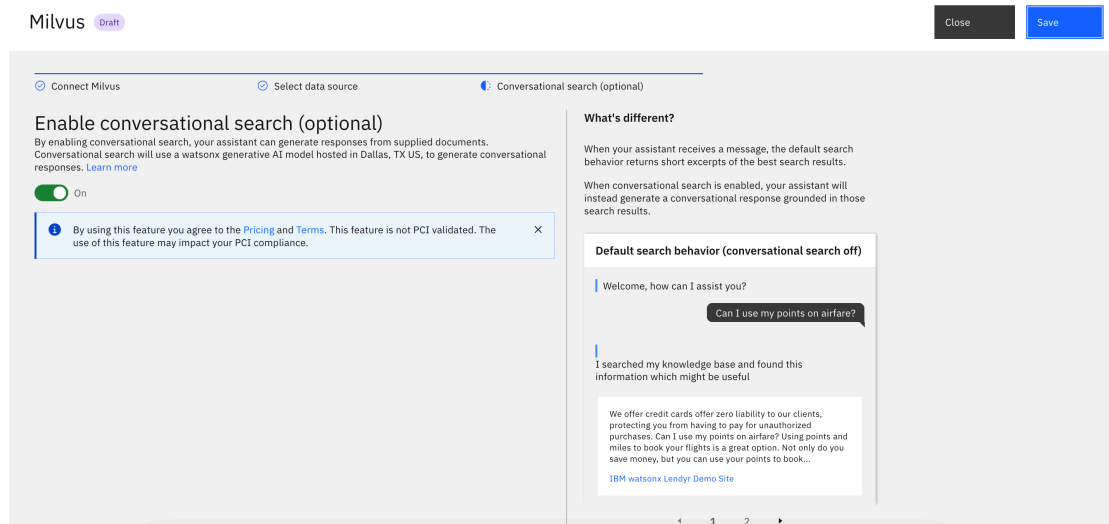
Body

text

URL (optional)

Select one

9.1.5 Click Next

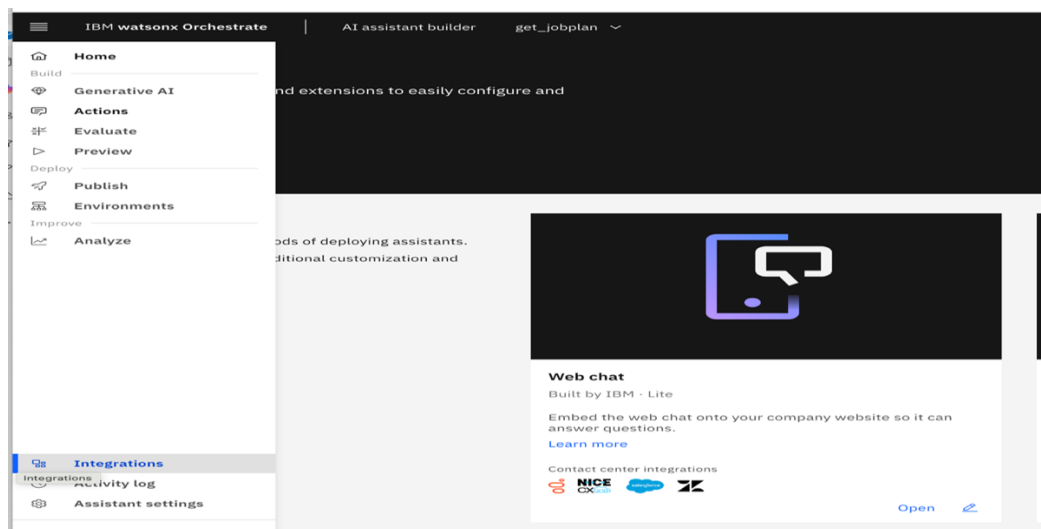


9.1.6 Click *Save* and *Close*.

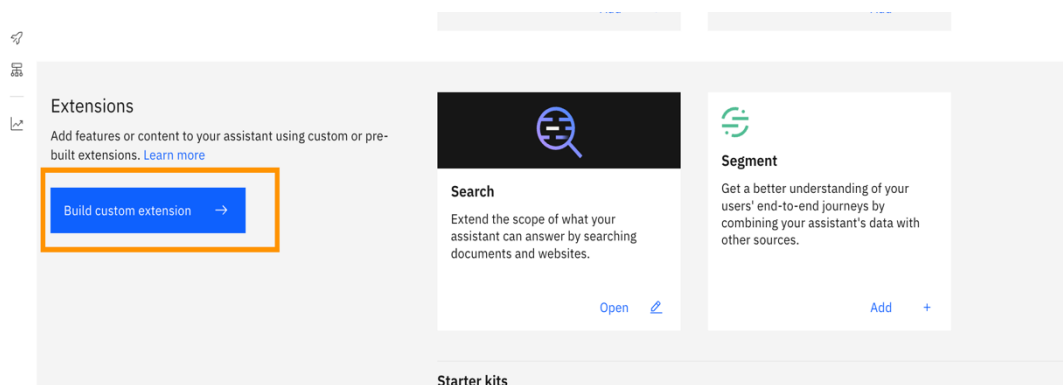
9.2 Maximo custom extension

9.2.1 Use **create_Jobplan.json** open Api file to add Maximo extension to the assistance

From Left Navigation, select *Integrations* option.



9.2.2 Click on *Build custom extension and click Next*



9.2.3 Enter Extension name as “maximo” and click next.

9.2.4 *Import open API* file “create_jobplan.json” and Click Next.

Custom extension

The screenshot shows the 'Import OpenAPI' step of the 'Custom extension' wizard. The progress bar at the top indicates the current step. The main heading is 'Import OpenAPI', followed by a description: 'Import an OpenAPI document in a .json format, describing the authentication and methods for your extension.' Below this is a dashed box with the text 'Drag and drop file here or click to upload'. At the bottom, a file named 'create_jobplan.json' is shown in a white box with a close button (X) on the right.

9.2.5 Review and *finish*.

The screenshot shows the 'Review extension' step of the 'Custom extension' wizard. The progress bar at the top indicates the current step. The main heading is 'Review extension', followed by a description: 'Review the servers and extension resources provided in the OpenAPI document.' Below this are two sections: 'Review servers' and 'Review operations'. The 'Review servers' section shows a table with one row: 'http://create-jobplan-maximo-flask.apps.itz-iyom01.infra01-lb.dal14.techzone.ibm.com'. The 'Review operations' section shows a table with one row: 'Create Job Plan' with a 'POST' method and a '/create-jobplan' resource. At the top right, there are 'Close' and 'Finish' buttons.

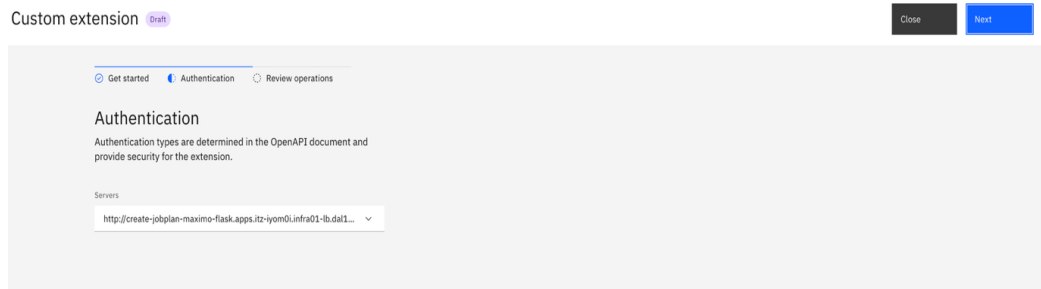
9.2.6 Then click on *Add*

The screenshot shows the 'Extensions' page of the 'Custom extension' wizard. The progress bar at the top indicates the current step. The main heading is 'Extensions', followed by a description: 'Add features or content to your assistant using custom or pre-built extensions. [Learn more](#)'. Below this is a blue button labeled 'Build custom extension' with a right arrow. To the right, there are three extension cards: 'Search', 'Segment', and 'maximo'. The 'maximo' card has an 'Add' button with a plus sign, which is highlighted with an orange box.

9.2.7 In Custom extension, click Next.

The screenshot shows the 'Get started' step of the 'Custom extension' wizard. The progress bar at the top indicates the current step. The main heading is 'Get started', followed by a description: 'How to add a custom extension to AI assistant builder:'. Below this are two numbered steps: '1. Use your credentials to authenticate the access necessary to add your extension.' and '2. Review the mapped variables table to ensure that the correct responses are available before adding the extension to your draft or live environment.' At the top right, there are 'Close' and 'Next' buttons.

9.2.8 Under *Authentication* you can see the Maximo URL (Same URL will be present in the open API file)



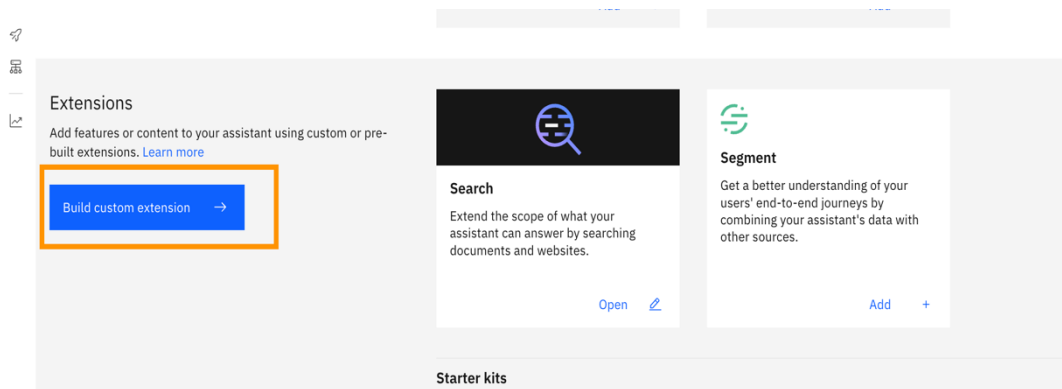
9.2.9 Click next & *finish*.



9.3 Watsonx AI integration

9.3.1 On the same Integrations page.

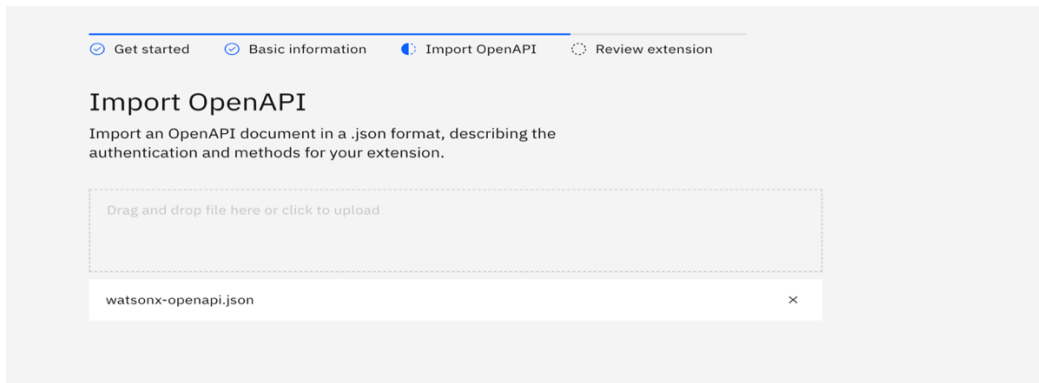
9.3.2 Click on *Build custom extension* again.



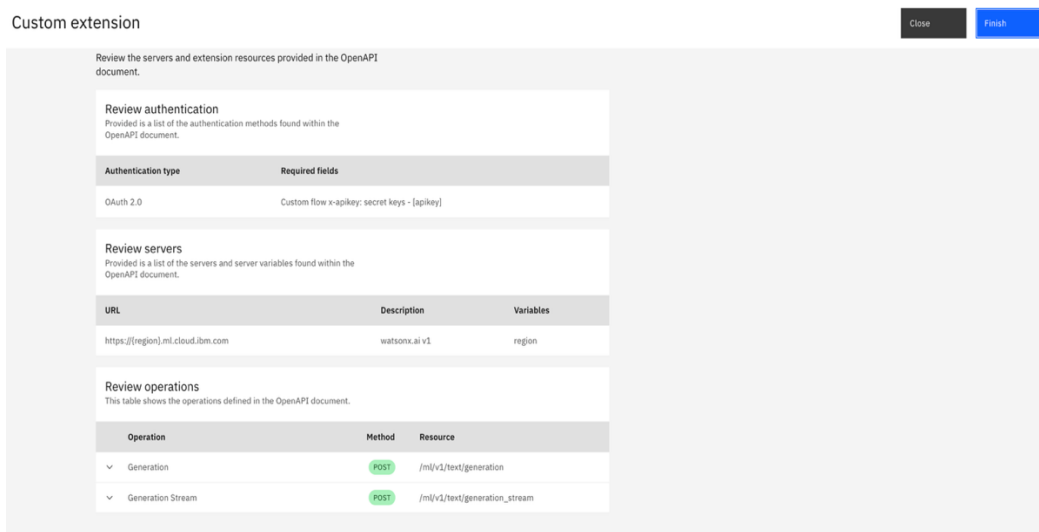
9.3.3 Enter Extension name as “watsonx ai”

9.3.4 In Import OpenAPI, select “[watsonx-openapi.json](#)” and click Next.

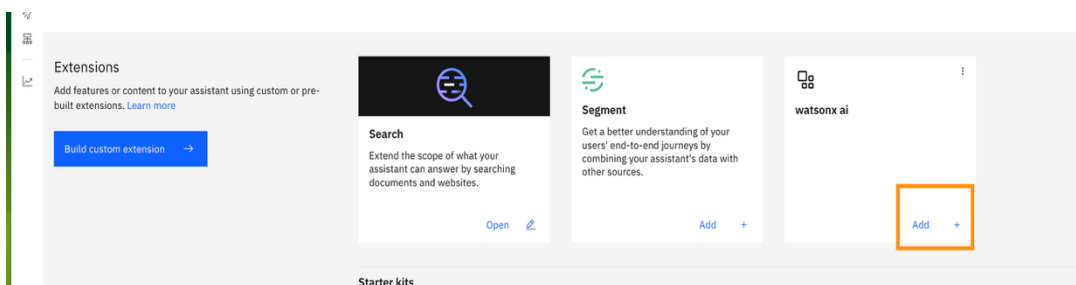
Custom extension



9.3.5 Review and click on *Finish*



9.3.6 Click on *Add*



9.3.7 In Authentication screen, Select/Provide below fields

- Authentication type: OAuth 2.0
- Grant type: Custom apikey
- Api key: provide watsonx.ai apikey
- Client authentication: sent as body

Custom extension Draft Close Next

Get started **Authentication** Review operations

Authentication

Authentication types are determined in the OpenAPI document and provide security for the extension.

Authentication type
OAuth 2.0

Grant type
Custom apikay

Custom Secrets
Apikay

Client authentication ⓘ
Send as Body

Header prefix ⓘ
Bearer

Servers
https://(region).ml.cloud.ibm.com

Generated URL: https://us-south.ml.cloud.ibm.com

Server variables
Enter values for your server variables to create a valid generated URL, for requests.

region
us-south

9.3.8 Click on next, Review & *Finish*

Custom extension Draft Close Finish

Get started Authentication **Review operations**

Review operations

This table shows the operations defined in the OpenAPI document.

Operation	Method	Resource
Generation	POST	/ml/v2/text/generation
Generation Stream	POST	/ml/v2/text/generation_stream

10. Import Actions

10.1 Use the file [actions.json](#) file

10.1.1 Select *Actions* form the left side panel

IBM watsonx Orchestrate | AI assistant builder milvus_data_source

Home

Build

Generative AI

Actions

Evaluate Actions

Preview

Deploy

Publish

Environments

Improve

Analyze

2025 10:20:21 AM GMT+5:30

as new watsonx generative AI features! [Learn more](#) about intelligence

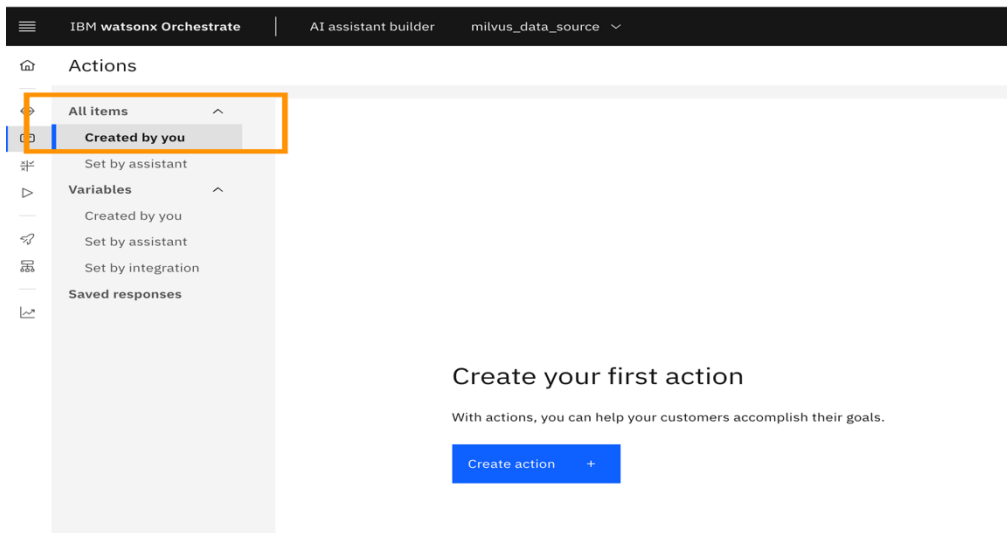
assistant

customize your assistant with these

Build actions

Enhance and improve

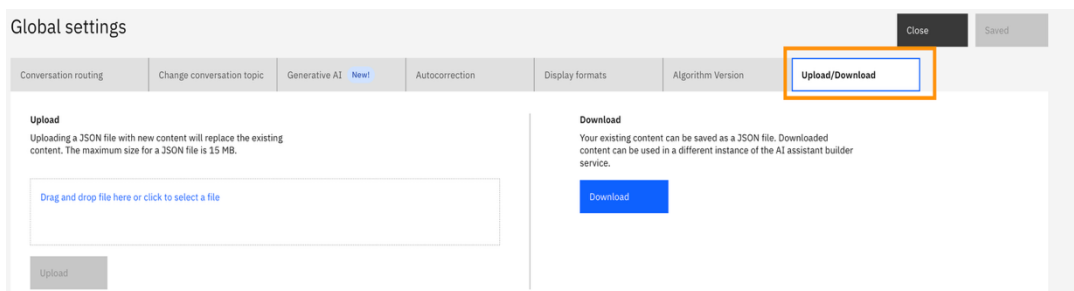
10.1.2 Click on *Created by You*



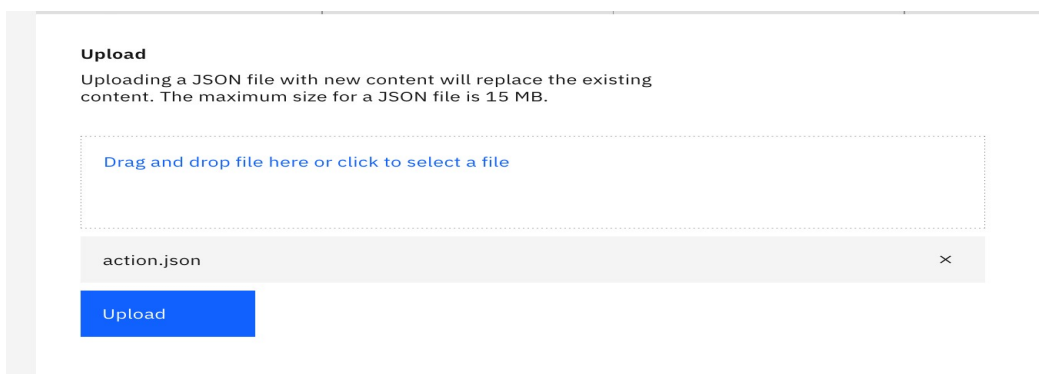
10.1.3 Select *Global settings* on right top corner



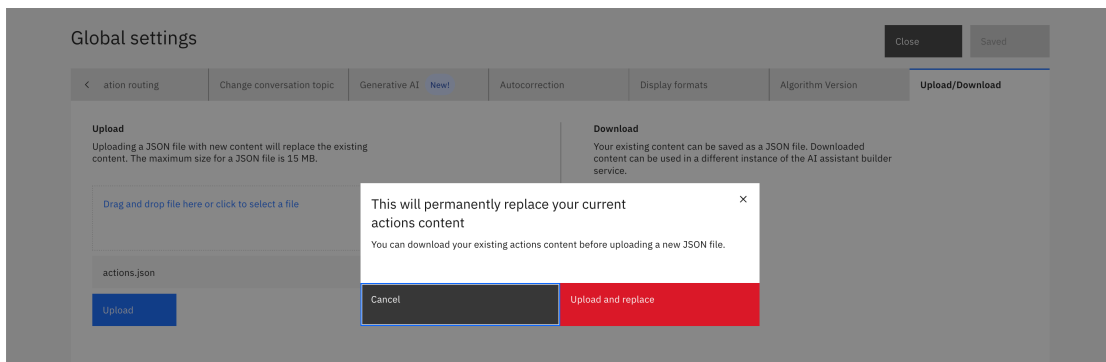
10.1.4 Navigate to *Upload/Download* option



10.1.5 Click on *Drag and drop file here* option & select “actions.json” file → Click on *Upload*



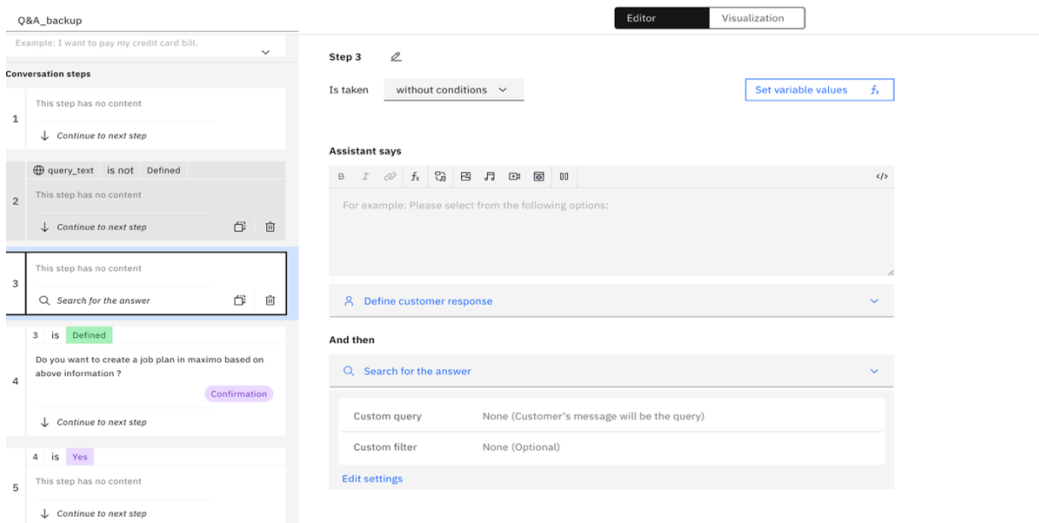
10.1.6 Select *Upload and replace* and Close the Global settings.



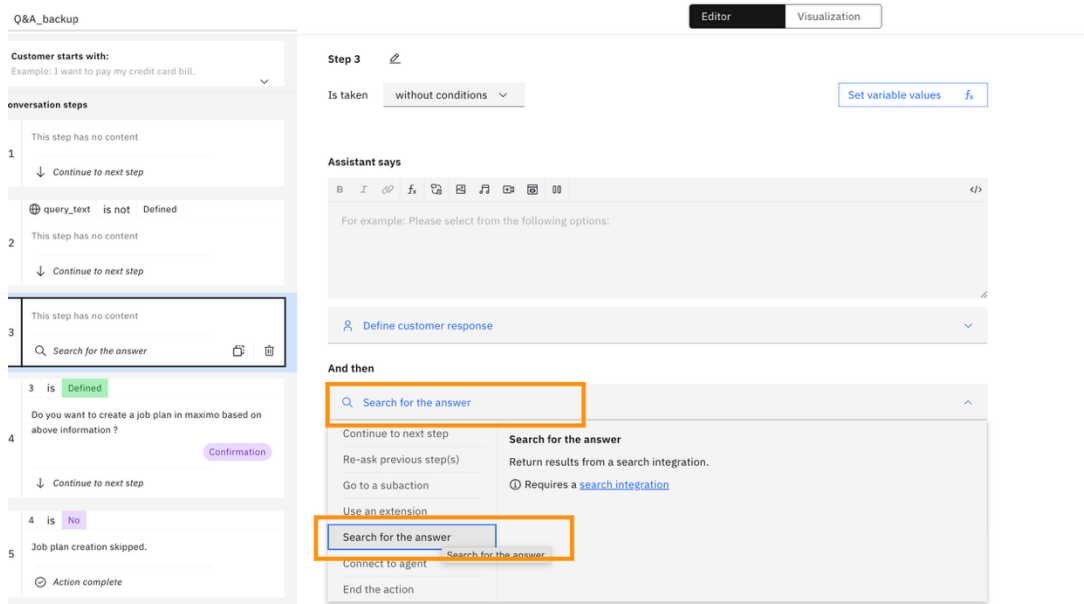
10.1.7 Click on *Q&A_backup* title



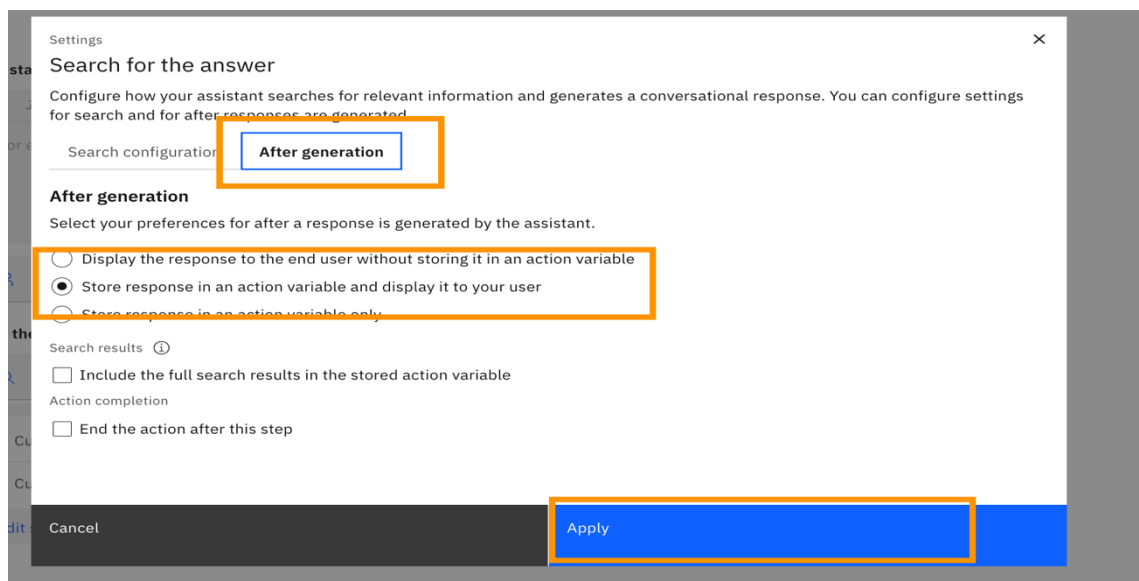
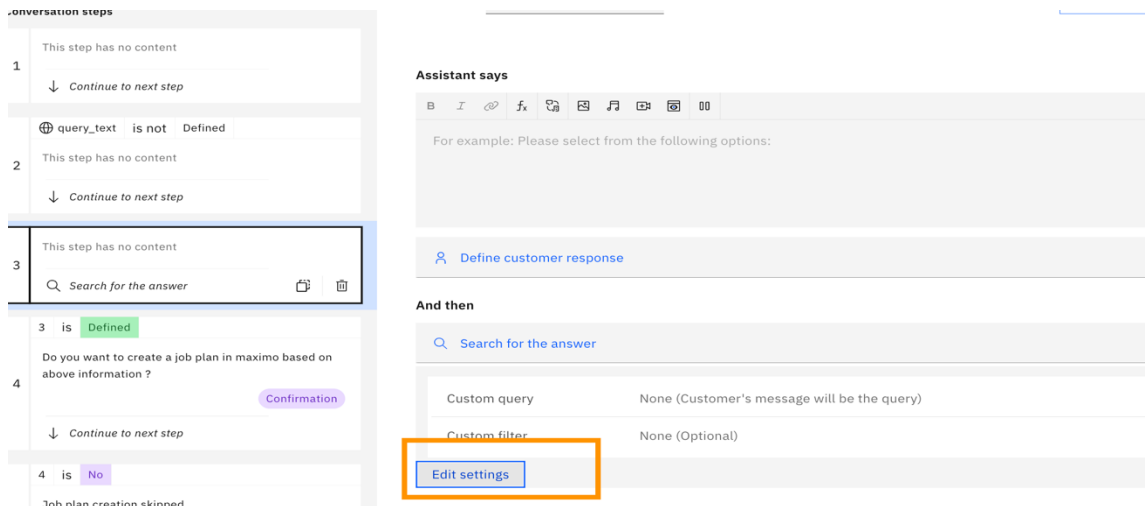
10.1.8 Click on Step3 → Make sure it is having *Search for the answer*



10.1.9 if not their select *Search for the answer*



10.1.10 Click on *Edit settings* → Click on *After generation* → select 2nd option (highlighted in image) → Click on *Apply*



10.1.11 Select step 7 and click on *Edit extension* and use watsonx.ai extension.

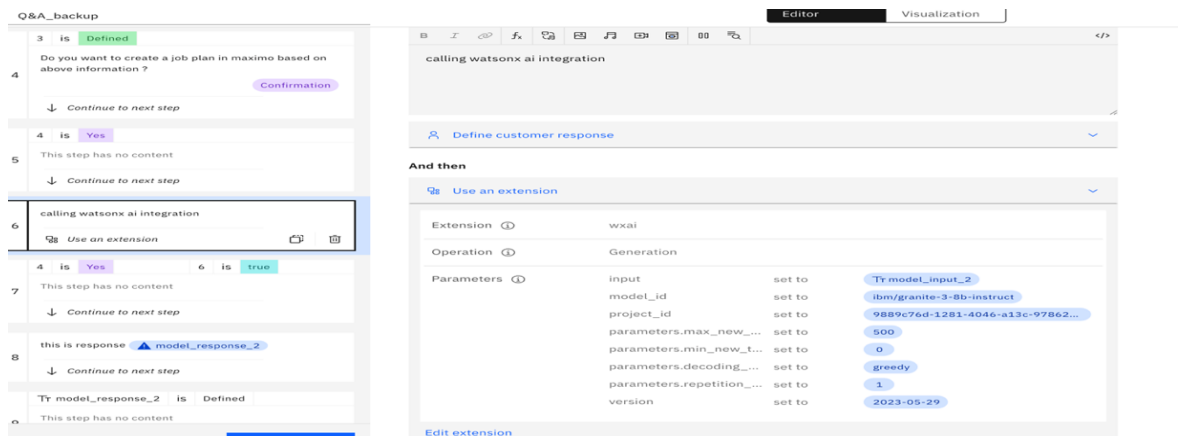
The screenshot shows the IBM Watson Assistant interface. On the left, a list of steps is visible, with step 7 highlighted. Step 7 is titled 'Processing your request through AI integration' and has a 'Use an extension' button. On the right, the 'Assistant says' section shows the message 'Processing your request through AI integration'. Below this, the 'And then' section shows the 'Use an extension' button. A red box highlights the 'Extension' dropdown menu, which is currently set to 'Extension 1'. A red error message 'Extension not fully configured' is displayed next to it. Below the dropdown is an 'Edit extension' button.

10.1.12 Update below fields → Click on *Apply*
Edit the details if it appears different to below options.

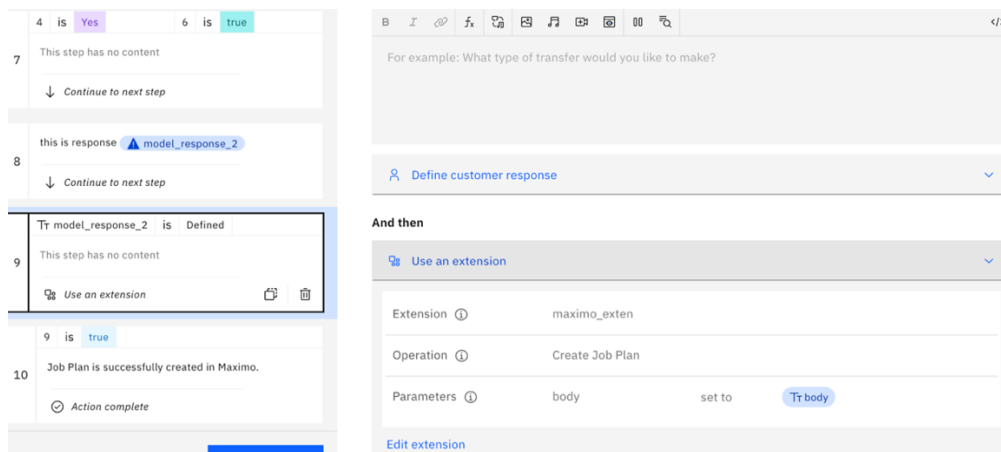
Field name	Fetching from
input	Previous step output (ex: model_input_2)
<i>Project_id</i>	<i>project id from watsonx.ai → step 4.7</i>
Model_id	<i>ibm/granite-3-8b-instruct</i>
Version	<i>2023-05-29</i>
Max_tokens	500 & above (based on generated answer)
Min_tokens	0

The screenshot shows the 'Extension setup' dialog box. It contains the following fields and options:

- Extension:** A dropdown menu with 'wxai' selected.
- Operation:** A dropdown menu with 'Generation' selected.
- Privacy:** A checkbox labeled 'Protect data returned from this extension' which is currently unchecked.
- Parameters:** A section with four 'Set' fields, each with a 'To' dropdown:
 - Set: 'Tr input' To: 'Tr model_input_2'
 - Set: 'Tr model_id' To: 'ibm/granite-3-8b-instruct'
 - Set: 'Tr project_id' To: '9889c76d-1281-4046-a13c-97862ba80591'
 - Set: 'Tr version' To: '2023-05-29'
- Optional parameters:** A section with an upward arrow icon.
- Buttons:** 'Cancel' and 'Apply' buttons at the bottom.

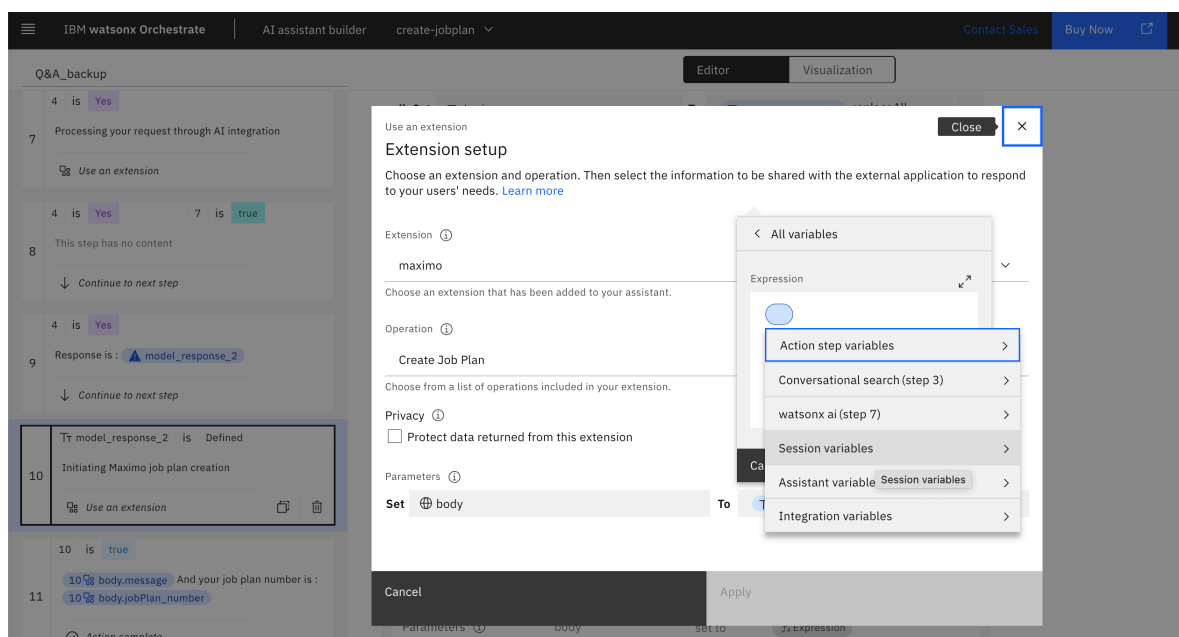


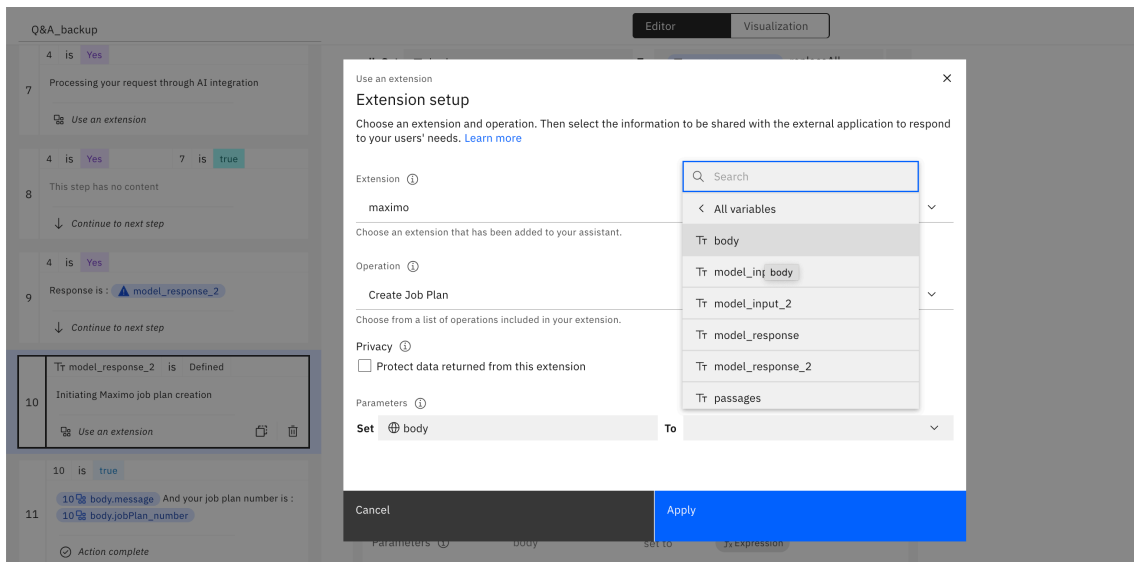
10.1.13 Select step 10 and click on *Edit extension* and use Maximo extension → provide *Parameters* value (previous step output in the actions (ex:body))



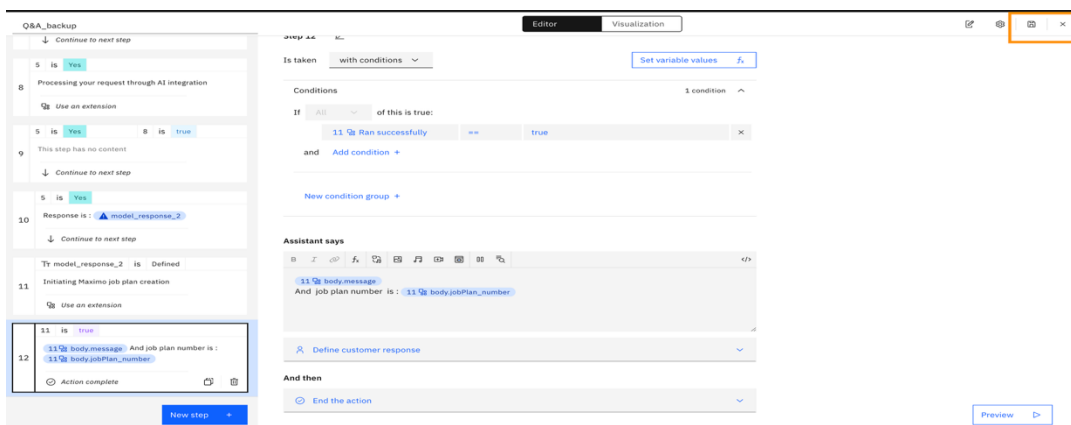
10.1.14 In Previous step, if Parameters is not set as Session variables (body) then follow this step.

- Click on Edit Extension
- Enter *Dollar (\$)* sign and select *Session variables* -> and select *body* -> apply

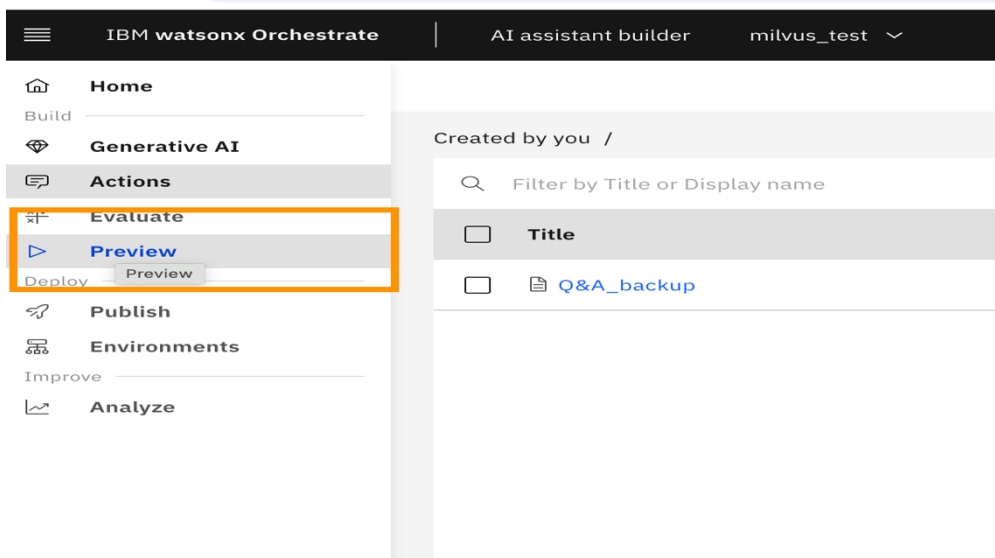




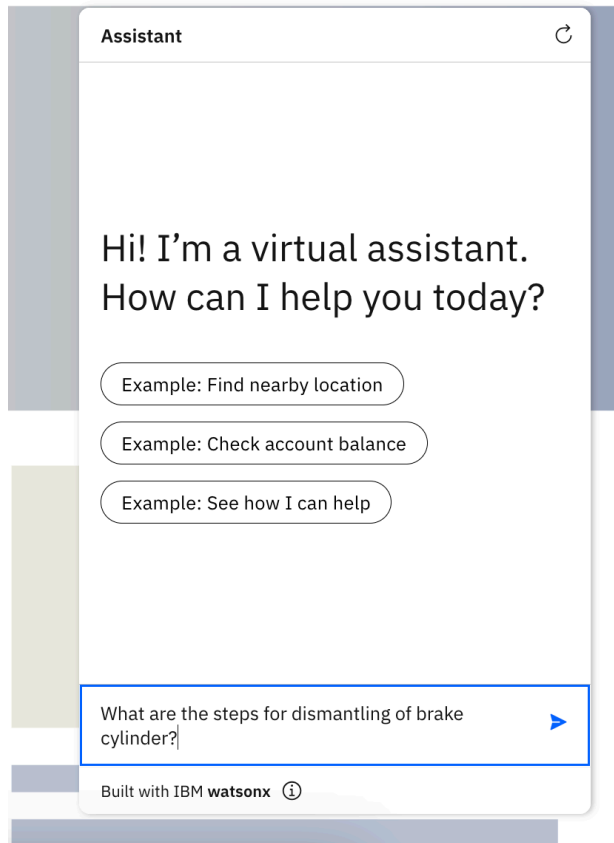
10.1.15 *Save* the actions → Click on *Close*



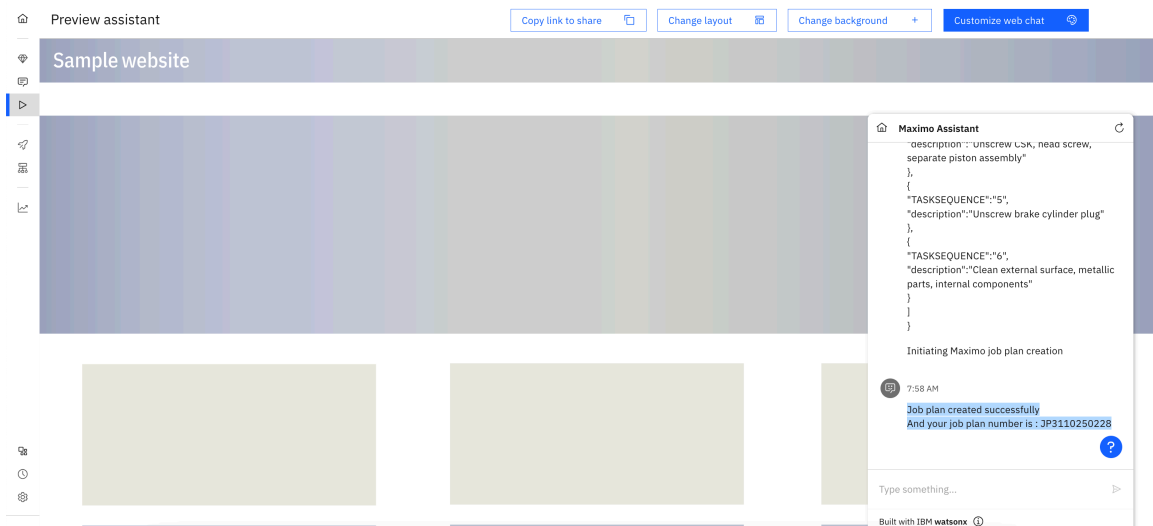
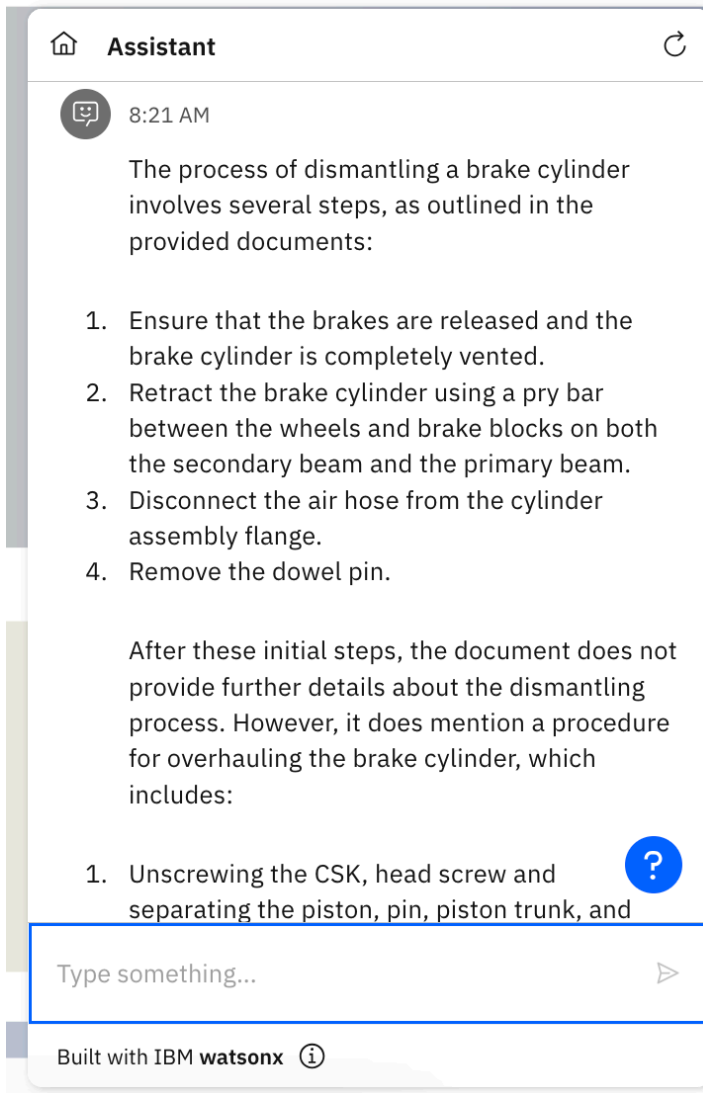
10.1.16 From *left* Navigation Click on *Preview*



10.1.17 Ask Question like - What are the steps for dismantling of brake cylinder?



10.1.18 Answer will be generated and displayed on the interface.



11. Conclusion/Summary

The automated creation of job plans in Maximo from asset maintenance manuals streamlines a traditionally manual, error-prone process. By leveraging IBM Watsonx technologies—including Watsonx.ai, Watsonx.data with Milvus, and Watson Assistant—organizations can extract structured maintenance tasks from unstructured manuals and convert them into standardized job plans within Maximo.

This AI-powered workflow enables:

- Digitization of OEM maintenance procedures
- Faster and more accurate job plan creation
- Intelligent retrieval of relevant content using vector search (Milvus)
- Seamless integration with Maximo through custom APIs

Ultimately, this solution enhances operational efficiency, improves asset reliability, ensures compliance with manufacturer guidelines, and reduces the time and effort required by maintenance teams.

By adopting this approach, customers can modernize their maintenance strategy and unlock real value from existing technical documentation.